UNIVERSITÉ Concordia UNIVERSITY CONTINUING EDUCATION

**MOD10: Introduction to Machine Learning**

# Project — Deliverable 1

*Dataset selection & problem framing*

**Due:** Lesson 2 (before class). **Groups:** 2–3 students. **Submit:** one PDF per group on Moodle (no code).

## Goal

The project runs across the whole term, one piece at a time. In this first deliverable, your group chooses a real-world dataset and frames a clear question you will answer with it. Everything you build in later weeks — preprocessing, models, evaluation — grows from the choices you make here, so choose something that genuinely interests your group.

## What to do

1. **Form your group** of 2–3 students.
2. **Choose a real-world dataset** (not a toy dataset built into a library). Good sources:
   UCI ML Repository · Kaggle · OpenML · Statistics Canada · Données Québec
3. **Frame your question** — what will you predict or discover? Decide your **problem type**: classification, regression, or clustering.
4. **Load it in Python** with `pd.read_csv(...)` and take a first look (`.head()`, `.shape`, `.info()`, `.isna().sum()`). This is just to check the file opens and to help you fill in the summary — **you do not submit any code for this deliverable.**
5. **Write a short summary** (1–2 pages) with the points listed below, and submit it as a PDF.

**You only submit the written summary as a PDF.** *There is no notebook or code to hand in for Deliverable 1 — the coding starts as a graded notebook in Deliverable 2.*

## What to include in your submission

- ☐ **Group members** (names).
- ☐ **Dataset**: name, source/link, and licence.
- ☐ **Size**: how many rows and columns.
- ☐ **Features**: a short description of the columns (numeric vs. categorical) and the target, if there is one.
- ☐ **Question**: what you want to answer, and why it matters.
- ☐ **Problem type**: classification, regression, or clustering — with a one-line justification.
- ☐ **Anticipated challenges**: e.g. missing data, class imbalance, size.

## How it will be assessed

Deliverable 1 is graded as part of the project's weekly deliverables. We look for:

- A **clear question** and an appropriate problem type, with a sensible reason.
- A **real, usable dataset** (right size, properly sourced and cited).
- Evidence you **looked at the data** — you can describe its columns and spot at least one challenge.

## Tips

- Tabular data with a mix of numeric and categorical columns works best.

- Aim for a few hundred to a few tens of thousands of rows — interesting, but small enough to run on your laptop or the lab machines.
- Avoid datasets with a single well-known “tutorial” answer (e.g. Titanic, Iris) unless you add a substantial twist.
- Do not use data that is sensitive or that you cannot legally/ethically use.

**Not sure if your dataset works?** Bring it to class or email me before you commit — it's much easier to adjust now than later.